

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Designed and tuned agentic AI workflows across multiple LLMs (GPT, Claude, Llama, Gemini), selecting models per use case for production quality and cost.
- Built a multi-agent system with the OpenAI Agents SDK and MCP tool-use for production workflows, with evaluation and guardrails around each agent step, handling millions of daily transactions at 99%+ uptime.
- Built GPU/HPC cluster infrastructure in Go and C++, scheduling jobs across Slurm and InfiniBand-networked nodes.
- Deployed a RAG pipeline on AWS SageMaker for real-time decision support at sub-200ms p99 latency.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Traced a recall regression in a production RAG pipeline to its chunking strategy and rebuilt retrieval with hybrid dense-sparse search and re-ranking, raising recall from 61% to 89% on a 2K-query eval set.
- Built a model observability framework tracking drift, latency, and eval regressions in production, enabling governance-aligned rollback workflows.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Built and operated 25+ production microservices in Java, C++, and SQL-backed distributed systems for 1M+ users, sustaining 99.9% uptime over 3 years on AWS (EC2, RDS, SQS).
- Standardized failure-handling contracts across 25+ financial services, reducing production escalations 60% in one quarter via consistent retry, timeout, and propagation semantics.

PROJECTS

Distributed KV Store / VPC Control Plane Simulator

github.com/ganeshhgupta/distributed-kv-store

- Built a Raft-consensus distributed storage engine from scratch (C++), implementing log compaction and linearizable reads under concurrent writes, a ground-up study of consensus algorithms.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- LangGraph multi-agent system automating production incident diagnosis via tool-use chains; routes alerts to specialized sub-agents and synthesizes root cause reports.

LLaMA 2 Fine-Tuning Pipeline for Financial Documents

github.com/ganeshhgupta/llama2-finance-finetuning

- Post-trained LLaMA 2 with LoRA and PEFT, running systematic hyperparameter optimization experiments to study capacity-performance tradeoffs.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Python, C++, Go, Java, TypeScript, SQL

Fundamentals: Data Structures & Algorithms, Consensus Algorithms, Applied Math, Distributed Systems

AI/ML: LLMs (GPT, Claude, Llama, Gemini), Agentic AI, RAG, LoRA/PEFT Fine-Tuning, Model Evaluation

Cloud & DevOps: AWS (EC2, S3, SageMaker), GCP (Cloud Run), Docker, Kubernetes, CI/CD, PostgreSQL

CERTIFICATIONS

AWS Certified Developer Associate | Google Cloud Professional Machine Learning Engineer